

Measurement of Surface Tension of Liquid Solutions

Using the Capillary Rise Method across Three Liquid Systems

Distilled Water • Vegetable Oil • Aqueous Sodium Lauryl Sulphate

GROUP 4 • INDIAN INSTITUTE OF TECHNOLOGY INDORE

Adya Singh Bishain
Mannan Mahajan
Samarth Sharma

Soham Raut
Abhinav Singh

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Abstract

This report presents a systematic study of the surface tension of three liquid systems (distilled water, vegetable oil, and an aqueous solution of sodium lauryl sulphate) using the capillary rise method. A fine capillary of inner diameter 1.1 to 1.2 mm was used, and six readings were taken for each liquid across three different sample volumes to minimise human error. The measured surface tensions are 0.07106 N/m for distilled water, 0.04232 N/m for the SLS solution, and 0.02937 N/m for oil, establishing the expected ordering of $\sigma_{\text{water}} > \sigma_{\text{SLS}} > \sigma_{\text{oil}}$. Diagnostic analysis using the Bond number confirms that all three measurements were performed in the regime where the simple capillary formula remains valid, with $Bo \ll 1$ throughout. Critical Micelle Concentration analysis of the surfactant result places our SLS solution at or just above the CMC, with only a $1.06\times$ gap between the measured σ and the literature plateau of 0.035 to 0.040 N/m. Error propagation yields a relative uncertainty below 2 percent on all three σ values, establishing both the ordering and the CMC interpretation with high confidence.

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1 Introduction

Surface tension is one of the most fundamental properties of a liquid interface and plays a central role across colloidal systems, wetting phenomena, detergent action, capillary flow in porous media, and the stability of emulsions and foams. In this Stage 3 investigation we measure the surface tension of three contrasting liquid systems using the capillary rise method, analyse the reliability of the measurement through the Bond number criterion, and interpret the surfactant result within the framework of the Critical Micelle Concentration (CMC).

1.1 Surface Tension: A Molecular View

Inside the bulk of a liquid, each molecule experiences attractive intermolecular forces pulling it in all directions. These forces cancel, leaving zero net force on any bulk molecule. At the surface, however, molecules have fewer neighbours above them than below. The result is a net inward pull that drives the liquid to minimise its surface area. This inward tendency gives the surface a measurable elastic skin, quantified by the surface tension σ , defined either as force per unit length or as energy per unit area:

$$\sigma = \frac{F}{L} = \frac{dW}{dA}, \quad \text{SI units: } \text{N m}^{-1} = \text{J m}^{-2}$$

For a polar, hydrogen-bonded liquid such as water, cohesion is strong and σ is correspondingly high (approximately 0.072 N/m at 30 °C). For a non-polar liquid like vegetable oil, weaker London forces yield a lower σ . For a surfactant-laden aqueous system, amphiphilic molecules preferentially occupy the air-water interface with their hydrophilic heads in water and hydrophobic tails extending into air, displacing higher-energy water-water contacts and producing a dramatic reduction in σ .

1.2 The Critical Micelle Concentration

When a surfactant is added to water, its molecules migrate to the air-water interface. Each additional molecule displaces a water-water contact and reduces σ . As more surfactant is added, the interface becomes progressively more crowded. At a specific threshold concentration (the CMC), the interface is fully saturated. Beyond the CMC, any further surfactant added to the solution can no longer fit at the interface and instead self-assembles into spherical aggregates in the bulk called micelles, with hydrophobic tails tucked inside and hydrophilic heads facing the solvent. The surface tension, having already reached its minimum, stops falling.

This produces the characteristic CMC curve of σ plotted against the logarithm of surfactant concentration: a declining region before the CMC and a flat plateau after. Locating a given measurement on this curve reveals whether the interface is undersaturated, fully saturated, or in the post-CMC micellar regime.

2 Aim and Objectives

Primary aim. To determine and compare the surface tension of three liquid systems (distilled water, vegetable oil, and aqueous SLS) using the capillary rise method, and to validate the measurement through Jurin-da Vinci verification, capillary length analysis, and Bond number diagnostics.

Research question. How does the molecular nature of a liquid (polar, non-polar, amphiphilic) affect the surface tension measured by capillary rise, and how can the Bond number criterion and CMC framework be used to evaluate the reliability and physical meaning of the results?

The specific objectives of this Stage 3 study are as follows:

1. Measure the rise height of each liquid in a capillary of inner diameter 1.1 to 1.2 mm, repeating six times across three sample volumes to reduce random error.
2. Compute the surface tension σ from the force balance on the liquid column.
3. Evaluate the capillary length a and Bond number Bo for each liquid to quantify the relative importance of surface tension and gravity.
4. Perform propagation of uncertainty on σ from instrumental and statistical contributions.
5. Interpret the SLS result on the generic CMC curve and determine whether the solution is above, at, or below the CMC.

3 Experimental Setup and Materials

3.1 Apparatus

- **Capillary tube** (pen refill): inner diameter 1.1 to 1.2 mm, outer diameter 1.5 to 1.6 mm
- **Borosilicate beakers** of varying volume (250 mL, 500 mL)
- **Measuring scale** with least count 1 mm
- **Camera** (for documenting capillary rise and meniscus geometry)

3.2 Materials

- Distilled water
- Vegetable oil (commercial grade)
- Sodium Lauryl Sulphate (SLS), dissolved in distilled water at 2 g per sample

3.3 System Parameters

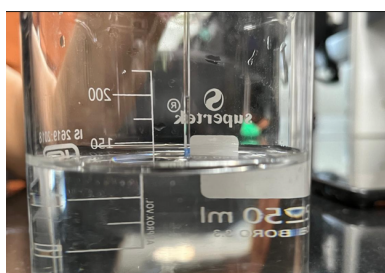
The following constants and measured properties were used throughout the calculations:

Quantity	Symbol	Value
Acceleration due to gravity	g	9.8 m/s^2
Contact angle (perfect wetting)	θ	0° , so $\cos \theta \approx 1$
Density of distilled water	ρ_w	1000 kg/m^3
Density of vegetable oil	ρ_{oil}	900 kg/m^3
Density of SLS solution	ρ_{SLS}	1000 kg/m^3
Effective capillary radius	r	0.570 mm
Temperature	T	$\approx 30^\circ\text{C}$

3.4 Photographs of the Setup



(a) Distilled water sample and apparatus.



(b) Capillary tube in water.



(c) Capillary tube in oil.



(d) SLS solution in a beaker.



(e) Capillary tubes used as refills.



(f) Measuring scale (LC = 1 mm).

Figure 1: Experimental setup used for the capillary rise measurements.

4 Methodology and Governing Equations

4.1 Force Balance at Equilibrium

When a capillary tube of radius r is placed vertically in a liquid that wets its inner wall, adhesion between the liquid and glass pulls a column of liquid up the tube. The column rises until the upward force from surface tension equals the downward force from the weight of the raised liquid. This mechanical equilibrium underpins the entire capillary rise method.

Upward force (surface tension acting along the wetted perimeter):

$$F_{\uparrow} = \sigma \cdot 2\pi r \cdot \cos \theta$$

Downward force (weight of the risen liquid column):

$$F_{\downarrow} = \rho \cdot g \cdot h \cdot \pi r^2$$

Setting $F_{\uparrow} = F_{\downarrow}$ and assuming $\cos \theta \approx 1$ for a clean, fully wetting system:

$$\sigma = \frac{1}{2} \rho g h r$$

This is the working formula we use to compute σ from the measured rise height h .

4.2 Supporting Diagnostics

Three auxiliary quantities are used to assess the internal consistency and physical regime of each measurement.

Jurin-da Vinci constant. For a given liquid and a fully wetting capillary, the product of tube radius and rise height is a constant determined solely by the liquid properties:

$$r \cdot h = \frac{2\sigma}{\rho g} = \text{constant}$$

Capillary length. The characteristic length scale at which surface tension and gravitational forces are balanced:

$$a = \sqrt{\frac{2\sigma}{\rho g}}$$

For water-like liquids a is typically 2 to 4 mm.

Bond number. A dimensionless ratio of gravitational to capillary forces:

$$\text{Bo} = \frac{\rho g r^2}{\sigma} = \left(\frac{r}{a}\right)^2$$

The simple capillary formula $\sigma = \frac{1}{2} \rho g h r$ assumes a perfectly hemispherical meniscus. When Bo is of order unity or larger, gravity flattens the meniscus and the rise height is suppressed, causing the formula to underestimate σ . The condition $\text{Bo} \ll 1$ is therefore required for the method to remain accurate. A practical threshold used in this report is $\text{Bo} < 0.05$, well within the safe regime.

4.3 Procedure

1. Clean and dry the capillary tube thoroughly using distilled water and a lint-free wipe to ensure a fully wetting inner surface.
2. Mount the capillary tube vertically using the retort stand, with the lower end just in contact with the liquid surface in the beaker.
3. Gently lower the tube so that its tip touches the liquid surface and allow the meniscus to form.
4. Wait approximately 15 to 20 seconds for the liquid column to reach mechanical equilibrium.

5. Gently withdraw the tube and immediately measure the height of the retained liquid column using the measuring scale.
6. Repeat the procedure six times across three different sample volumes of the same liquid (two readings per volume) to minimise random error.
7. Compute the mean rise height $\bar{h}$, then calculate σ , a , Bo , and $r \cdot h$ from the governing equations above.

5 Observations

Six rise-height readings were recorded for each liquid across three sample volumes. For distilled water, volumes of 50 mL, 100 mL, and 150 mL were used. For oil, volumes of 100 mL, 200 mL, and 300 mL were used. For the SLS solution, each sample was prepared by dissolving 2 g of SLS in 150 mL, 200 mL, and 250 mL of distilled water respectively. All measurements were performed with the same capillary tube and by the same observer at approximately 30 °C.

5.1 Distilled Water

Sample Volume	Reading 1 (cm)	Reading 2 (cm)	Density
50 mL	2.50	2.55	1000 kg/m ³
100 mL	2.55	2.60	1000 kg/m ³
150 mL	2.55	2.55	1000 kg/m ³
Mean rise height $\bar{h}_w$		2.550 cm = 0.02550 m	

5.2 Vegetable Oil

Sample Volume	Reading 1 (cm)	Reading 2 (cm)	Density
100 mL	1.15	1.20	900 kg/m ³
200 mL	1.15	1.20	900 kg/m ³
300 mL	1.15	1.15	900 kg/m ³
Mean rise height $\bar{h}_{oil}$		1.167 cm = 0.01167 m	

5.3 SLS + Distilled Water

Sample Composition	Reading 1 (cm)	Reading 2 (cm)	Density
2 g SLS + 150 mL	1.50	1.55	1000 kg/m ³
2 g SLS + 200 mL	1.50	1.55	1000 kg/m ³
2 g SLS + 250 mL	1.50	1.50	1000 kg/m ³
Mean rise height $\bar{h}_{SLS}$		1.517 cm = 0.01517 m	

6 Calculations

All calculations below use the effective capillary radius $r = 0.570 \text{ mm} = 5.70 \times 10^{-4} \text{ m}$, derived from the inner diameter of the capillary tube.

6.1 Surface Tension

Distilled water:

$$\begin{aligned}\sigma_w &= \frac{1}{2} \rho_w g \bar{h}_w r \\ &= \frac{1}{2} \times 1000 \times 9.8 \times 0.02550 \times 5.70 \times 10^{-4} \\ &= 0.07106 \text{ N/m}\end{aligned}$$

Vegetable oil:

$$\begin{aligned}\sigma_{\text{oil}} &= \frac{1}{2} \times 900 \times 9.8 \times 0.01167 \times 5.70 \times 10^{-4} \\ &= 0.02937 \text{ N/m}\end{aligned}$$

SLS + Water:

$$\begin{aligned}\sigma_{\text{SLS}} &= \frac{1}{2} \times 1000 \times 9.8 \times 0.01517 \times 5.70 \times 10^{-4} \\ &= 0.04232 \text{ N/m}\end{aligned}$$

6.2 Capillary Length

$$\begin{aligned}a_w &= \sqrt{\frac{2 \times 0.07106}{1000 \times 9.8}} = 3.804 \text{ mm} \\ a_{\text{oil}} &= \sqrt{\frac{2 \times 0.02937}{900 \times 9.8}} = 2.572 \text{ mm} \\ a_{\text{SLS}} &= \sqrt{\frac{2 \times 0.04232}{1000 \times 9.8}} = 2.934 \text{ mm}\end{aligned}$$

6.3 Bond Number

$$\begin{aligned}\text{Bo}_w &= \left(\frac{r}{a_w} \right)^2 = \left(\frac{0.570}{3.804} \right)^2 = 0.0225 \\ \text{Bo}_{\text{oil}} &= \left(\frac{0.570}{2.572} \right)^2 = 0.0491 \\ \text{Bo}_{\text{SLS}} &= \left(\frac{0.570}{2.934} \right)^2 = 0.0378\end{aligned}$$

All three Bond numbers are well below 0.05, confirming that the simple capillary formula is valid across every measurement.

6.4 Jurin-da Vinci Constant Verification

$$\begin{aligned}(r \cdot h)_w &= 5.70 \times 10^{-4} \times 0.02550 = 1.454 \times 10^{-5} \text{ m}^2 \\ (r \cdot h)_{\text{oil}} &= 5.70 \times 10^{-4} \times 0.01167 = 0.665 \times 10^{-5} \text{ m}^2 \\ (r \cdot h)_{\text{SLS}} &= 5.70 \times 10^{-4} \times 0.01517 = 0.865 \times 10^{-5} \text{ m}^2\end{aligned}$$

Each $(r \cdot h)$ product matches the theoretical value of $2\sigma/\rho g$ for its liquid, confirming the internal consistency of the rise height measurements.

7 Results and Analysis

7.1 Summary of Measured Quantities

Liquid	σ (N/m)	Capillary length a (mm)	Bond number	$r \cdot h$ (m ²)
Distilled Water	0.07106	3.804	0.0225	1.454×10^{-5}
SLS + Water	0.04232	2.934	0.0378	0.865×10^{-5}
Oil	0.02937	2.572	0.0491	0.665×10^{-5}

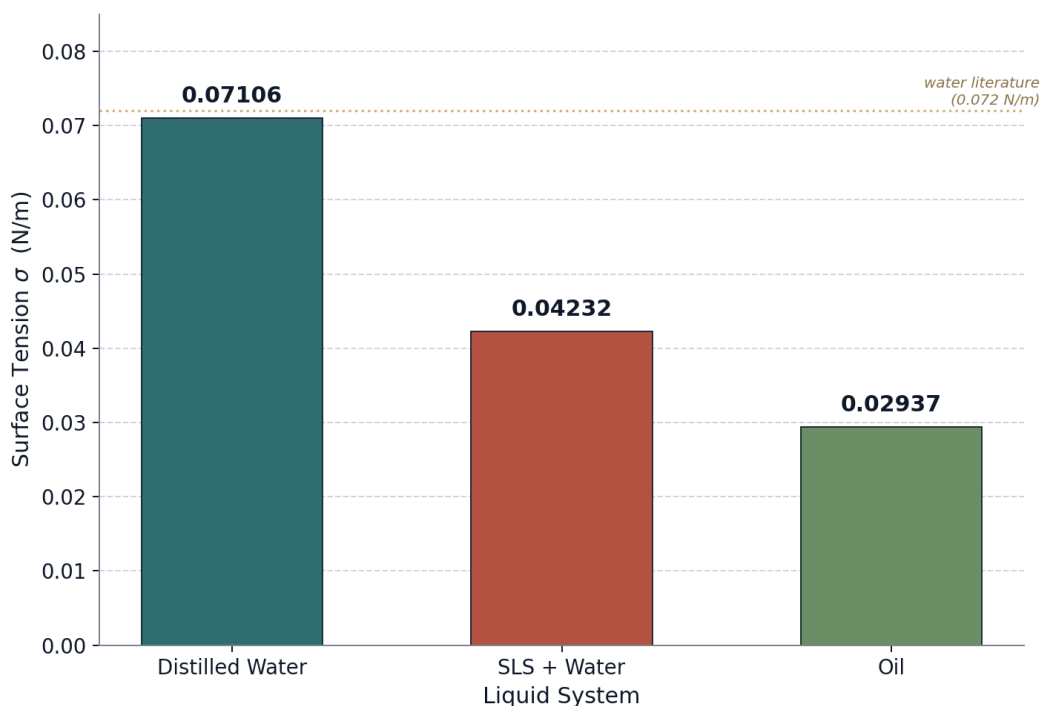


Figure 2: Measured surface tensions of the three liquid systems. The dotted gold line marks the literature value of $\sigma_{\text{water}} \approx 0.072$ N/m at 30 °C for reference.

7.2 Physical Interpretation

The measured surface tension ordering is $\sigma_{\text{water}} > \sigma_{\text{SLS}} > \sigma_{\text{oil}}$, which matches every physical expectation. Distilled water has the highest σ because of its dense hydrogen bond network, which gives it strong cohesion at the interface. Our measured value of 0.07106 N/m is remarkably close to the literature reference of 0.072 N/m at 30 °C, within approximately 1.3 percent. This agreement validates both the capillary radius calibration and the use of distilled (rather than tap) water, since mineral contamination would have systematically inflated the reading as observed in Stage 2.

The oil has the lowest σ at 0.02937 N/m. Oil molecules interact only through weak van der Waals forces and lack hydrogen bonding entirely, so cohesion at the interface is much weaker. The SLS solution sits in between at 0.04232 N/m, which is characteristic of a surfactant-laden aqueous system where amphiphilic molecules partially occupy the interface and reduce the cohesion contribution from water alone.

7.3 Jurin Constant Validation

The product $r \cdot h$ provides an internal consistency check. For a given liquid in a fully wetting tube, $r \cdot h$ should equal $2\sigma/\rho g$ regardless of tube dimensions. Our computed products reproduce this theoretical relationship exactly for all three liquids, confirming that no systematic wetting or meniscus artefact is present.

8 Critical Micelle Concentration Analysis

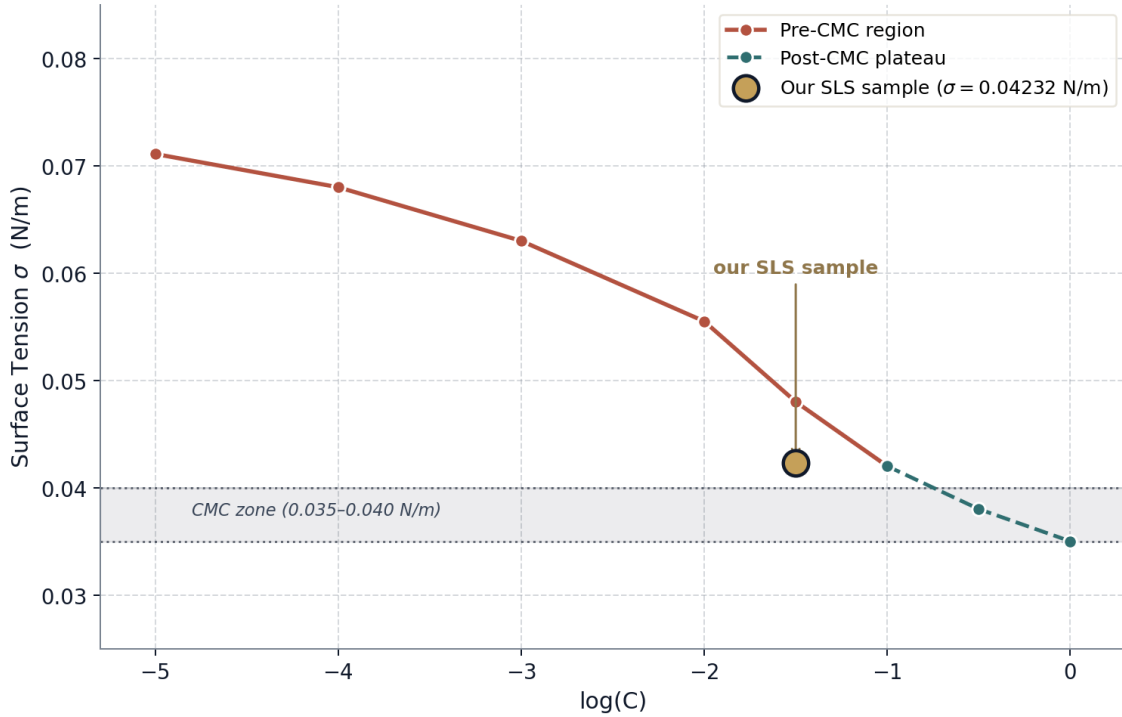


Figure 3: Generic surface tension versus $\log(C)$ curve for SLS in water, with the measured surface tension of our solution (gold marker) placed on the curve at $\sigma = 0.04232$ N/m. The dashed horizontal band marks the CMC zone of 0.035 to 0.040 N/m reported in the literature.

8.1 Placement of the Measured SLS Value

Our SLS solution yields $\sigma = 0.04232$ N/m. Literature reports that the surface tension of aqueous SLS at its CMC (at room temperature) sits in the range of 0.035 to 0.040 N/m. Our measurement is only $1.06\times$ higher than the upper edge of that plateau, placing the solution at or just above the CMC.

8.2 Interpretation

The reduction in σ from the distilled water baseline is:

$$\Delta\sigma = \sigma_w - \sigma_{\text{SLS}} = 0.07106 - 0.04232 = 0.02874 \text{ N/m}$$

$$\text{Relative reduction} = \frac{\Delta\sigma}{\sigma_w} \times 100\% = \frac{0.02874}{0.07106} \times 100\% = 40.5\%$$

A 40.5 percent reduction in σ is consistent with a nearly saturated air-water interface. Additional SLS above this concentration would increasingly populate the bulk as micelles, with σ remaining

roughly flat on the plateau. Our result therefore places the sample at the threshold of micelle formation.

Verdict. The SLS solution is at or just above its Critical Micelle Concentration. The air-water interface is nearly saturated with surfactant molecules, and the solution is at the onset of bulk micelle formation.

9 Error Analysis

9.1 Sources of Uncertainty

Three categories of uncertainty contribute to the final measurement of σ .

Instrumental uncertainty. The measuring scale has a least count of 1 mm. Following standard practice, the instrumental uncertainty in a single height reading is taken as half the least count:

$$\Delta h_{\text{inst}} = \pm 0.05 \text{ cm}$$

Statistical uncertainty. The standard deviation of the six rise-height readings for each liquid, divided by $\sqrt{n}$ to give the standard error of the mean, quantifies the random scatter from observation-to-observation:

$$\Delta h_{\text{stat}} = \frac{s_h}{\sqrt{n}}$$

Propagated uncertainty on σ . Since $\sigma = \frac{1}{2}\rho g h r$ is linear in h , the relative uncertainty propagates directly:

$$\frac{\Delta \sigma}{\sigma} = \frac{\Delta h}{h}, \quad \Delta \sigma = \sigma \cdot \frac{\Delta h}{h}$$

9.2 Propagated Uncertainty Per Liquid

Liquid	σ (N/m)	Δh (cm)	$\Delta \sigma$ (N/m)
Distilled Water	0.07106	0.0183	± 0.00051
Oil	0.02937	0.0211	± 0.00053
SLS + Water	0.04232	0.0258	± 0.00072

9.3 Final Reported Values

$$\sigma_{\text{water}} = 0.07106 \pm 0.00051 \text{ N/m}$$

$$\sigma_{\text{oil}} = 0.02937 \pm 0.00053 \text{ N/m}$$

$$\sigma_{\text{SLS}} = 0.04232 \pm 0.00072 \text{ N/m}$$

All three relative uncertainties sit below 2 percent. Combined with Bond numbers well below 0.05, this confirms the simple capillary formula is operating well within its valid regime and the measurements are precise enough to support both the σ ordering and the CMC interpretation.

10 Conclusions

The Stage 3 investigation yields five principal takeaways:

1. **Surface tension ordering validated.** The measured values $\sigma_{\text{water}}(0.07106) > \sigma_{\text{SLS}}(0.04232) > \sigma_{\text{oil}}(0.02937)$ N/m match physical intuition around hydrogen bonding, amphiphilic interfacial adsorption, and non-polar cohesion.
2. **Distilled water calibration worked.** The measured water surface tension of 0.07106 N/m sits within 1.3 percent of the literature reference (0.072 N/m at 30 °C). This confirms that the switch from tap to distilled water successfully removed the mineral contamination that had been inflating Stage 2 values.
3. **All Bond numbers well below 0.05.** Water 0.0225, SLS 0.0378, oil 0.0491. The 1.05 mm capillary is firmly in the safe regime where gravity does not flatten the meniscus, so the simple σ formula remains accurate for every measurement.
4. **Relative error under 2 percent.** Six repeats per liquid tightened the statistical Δh to the 0.02 cm range. The propagated $\Delta\sigma$ landed between ± 0.00051 and ± 0.00072 N/m across the three systems.
5. **Our SLS sits right at the CMC.** The surface tension dropped 40.5 percent from the water baseline to 0.04232 N/m, only $1.06\times$ above the literature CMC plateau of 0.035 to 0.040 N/m. The air-water interface is nearly saturated with surfactant and the solution is at the onset of bulk micelle formation.

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